

**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI,
K K BIRLA GOA CAMPUS**

No electronic devices allowed, single A4 sheet (both sides) is allowed.

Subject Name: BITS F464 - Machine Learning,

Date: 06 May 2024

Examiner Name: Aditya Challa

Maximum Marks: 40

Duration: 2.5 hours (10:00 AM - 12:30 PM)

Instructions

- **Attempt all questions.**
- **Marks corresponding to each question is highlighted in bold within square braces at the end of the question.**
- **You are allowed to a single A4 sheet both sides with any content you want. Make sure to write your BITS ID/Name on the sheet.**
- **In case of ambiguities in any of the questions, clearly state your assumptions and attempt the question(s).**
- **You should write the final answers in the space provided on the question paper. Answers not written in the space provided will not be evaluated.**
- **You can use your answer sheet for rough work.**
- **Return both the question paper and the answer-sheet at the end of the examination.**

Please fill the following details:

Name: BITS ID: BITS Email-ID: (Used for google classroom)**Problem 1**

1. Suppose we have a domain knowledge that the ground-truth function is a finite sum of the form $f(t) = \sum_{i=1}^p a_i \cos(2\pi\omega_i t)$, where $a_i \in \mathbb{R}$ is fixed but unknown, but know that ω_i belongs to the set of integers between 1 and $K - \{1, 2, \dots, K\}$. So, sampling from this function will give observations - $\{(t_i, y_i = f(t_i))\}_{i=1}^n$. Assume t_i is uniformly sampled from $[-1, 1]$.

Using the domain knowledge, let us construct the hypothesis class $\mathcal{H}_c$ (indexed by c) as follows - $\mathcal{H}_c = \{h(t) = \sum_{i=1}^c a_i \cos(2\pi\omega_i t)\}$, where a_i is some real number and $\omega_i \in \{1, 2, \dots, K\}$.

Recall that we state $\text{Complexity}(\mathcal{H}_1) \leq \text{Complexity}(\mathcal{H}_2)$ only if $\mathcal{H}_1 \subseteq \mathcal{H}_2$.

- 1(i) Select ALL the correct statements from the following. (Be careful about $<$ vs $\leq$).
- (a) If $c_1 < c_2$, then $\text{Complexity}(\mathcal{H}_{c_1}) < \text{Complexity}(\mathcal{H}_{c_2})$
 - (b) If $c_1 < c_2$, then $\text{Complexity}(\mathcal{H}_{c_1}) \leq \text{Complexity}(\mathcal{H}_{c_2})$
 - (c) If $c_1 \leq c_2$, then $\text{Complexity}(\mathcal{H}_{c_1}) < \text{Complexity}(\mathcal{H}_{c_2})$
 - (d) If $c_1 \leq c_2$, then $\text{Complexity}(\mathcal{H}_{c_1}) \leq \text{Complexity}(\mathcal{H}_{c_2})$
- 1(ii) Assume that there is a *known* procedure which identifies the optimal hypothesis from the class $\mathcal{H}_c$, i.e minimizes the mean squared error on the sample. Which of the following statements are true? Hint: Nyquist-Shannon theorem states - If a function $x(t)$ contains no frequencies higher than B hertz, then it can be completely determined from its ordinates at a sequence of points spaced less than $(1/2B)$ seconds apart.

- (a) Fitting the optimal hypothesis from $\mathcal{H}_K$ will give zero error, assuming large enough sample size n .
- (b) Fitting the optimal hypothesis from $\mathcal{H}_K$ will give zero error, irrespective of the sample size n .
- (c) If the sample consists of the values at $t = \{-1, -1 + 1/3K, -1 + 2/3K, \dots, 1\}$, then the error is 0.
- (d) If the sample consists of the values at $t = \{-1, -1 + 1/3K, -1 + 2/3K, \dots, 1\}$, then the variance of our estimate is 0.
- 1(iii) One way to look at the problem is as a *regression* problem. Since, K is known, we can generate K features by transforming t_i to $(\cos(2\pi t_i), \cos(2\pi 2t_i), \dots, \cos(2\pi K t_i))$. So we now have K features and we need to identify the values of a_i , and can use the linear regression model to do so.
- Which of the following statements are true?
- (a) Since there are K unknowns - $\{a_i\}$, we need at least K distinct (w.r.t t) data points to identify the values of a_i .
- (b) Since there are K unknowns - $\{a_i\}$, we need at most K distinct (w.r.t t) data points to identify the values of a_i .

Answer Here:

1(i)

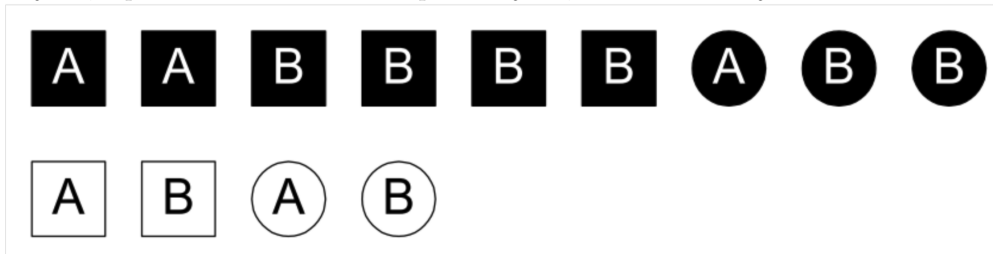
1(ii)

1(iii)

[4 + 4 + 2 = 10 Marks]

Problem 2

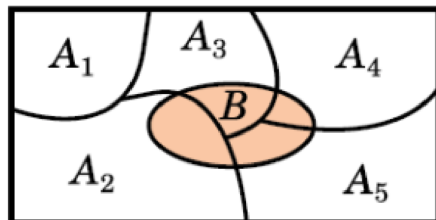
2. In all questions, there may be 1 or more correct answers, or no correct answer. In the former case, full marks are given if all correct answers and none of the incorrect answers are selected. In the latter case, full marks are given if only the “None of these” option is selected and none of the other options are selected.
- 2(i) In the figure below, let *Black* denote the set of black objects, *White* denote the set of white objects, *Square* denote the set of square objects, *A* the set of objects with an “A” and so on.



Assuming all objects are equally likely which of the following are true:

- (a) $P(A) = 5/13$
- (b) $P(A|Square) = 3/8$
- (c) *A* and *Square* are independent
- (d) *A* and *Black* are independent
- (e) *A* and *Square* are conditionally independent given *Black*
- (f) *A* and *Square* conditionally independent given *White*
- (g) $P(Black|A) < P(White|A)$.

- (h) None of these
- 2(ii) The sample space Ω is divided into sets A_1 to A_5 , which partly overlap with set B as shown below.



For any set S , S^c denotes the complementary set. Which of the following are correct?

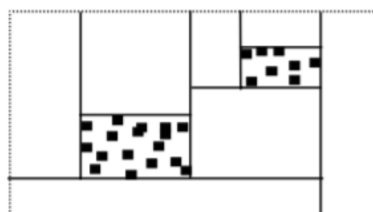
- (a) $P(B) = \sum_{i=1}^5 P(B \cap A_i)$
- (b) $P(B) = P(B|A_1) + P(B|A_2) + P(B|A_3) + P(B|A_4) + P(B|A_5)$
- (c) $P(B) = P(B|A_1)P(A_1) + P(B|A_1^c)P(A_1^c)$
- (d) None of these
- 2(iii) You are given a sample of n observations data $x_1, x_2, \dots, x_n$ from a binomial distribution with parameters N and p (that is, observation in sample i is the number of successes in that sample). Assume N and p are known, $q = 1 - p$, and L denotes the likelihood of the data observed. Which of the following are true:
- (a) $L = \prod_i \binom{N}{x_i} p^{x_i} q^{(N-x_i)}$
- (b) $L = \prod_i p^{x_i} q^{(N-x_i)}$
- (c) $L \propto C p^{\sum_i x_i} q^{(Nn - \sum_i x_i)}$
- (d) None of these
- 2(iv) You are given a sample of n observations data $x_1, x_2, \dots, x_n$ from a binomial distribution with parameters N and p (that is, observation in sample i is the number of successes in that sample). Assume N is known and, $q = 1 - p$, and p_{MLE} denotes the maximum-likelihood estimate for p . Which of the following are true:
- (a) $p_{MLE} = \frac{1}{N} \sum_i x_i$
- (b) $p_{MLE} = \frac{1}{Nn} \sum_i x_i$
- (c) $p_{MLE} = \frac{1}{n} \sum_i x_i$
- (d) None of these
- 2(v) The table below summarises the number of times tokens α , β , γ occur in 8 documents. You are asked to model the data by Boolean-valued r.v.'s A, B, C denoting the occurrence or non-occurrence of tokens α , β and γ (anywhere) in a document. The documents themselves are labelled + ("interesting") or - ("not interesting"). four in the positive class, and four negative.

Document No.	α	β	γ	Label
1	2	0	4	+
2	0	3	3	+
3	3	0	0	+
4	0	0	2	+
5	0	0	0	-
6	3	0	0	-
7	4	3	0	-
8	4	0	0	-

Assume: (a) the label is an occurrence of a random variable Y ; (b) that variables A, B, C are conditionally independent of each other given Y ; and (c) $Beta(1, 1)$ priors when calculating estimates of probabilities of all 2-valued r.v.'s.

An unlabelled document d contains 2 occurrences of token α , 3 occurrences of token γ and 0 occurrences of token β . Which of the following are true:

- (a) $P(Y = +|d) \in [0, 0.05]$
 (b) $P(Y = +|d) \in [0.025, 0.075]$
 (c) $P(Y = +|d) \in [0.05, 0.10]$
 (d) None of these
- 2(vi) A Gaussian Naive Bayes Classifier is a generative model $P(X, Y)$ that makes the following assumptions: (1) Y is a Bernoulli r.v. with parameter $\pi = P(Y = 1)$; (2) $X = (X_1, X_2, \dots, X_n)$, where each X_i is a continuous r.v.; (3) The class-conditional density of each of the X_i i.e. $P(X_i|Y = y_k)$ is defined by a Gaussian distribution $N(\mu_{ik}, \sigma_i)$; and (4) for $i \neq j$, the $X_{i,j}$ are conditional independent of each other, given Y . Starting from Bayes Rule, and the information above, it follows that $P(Y = 1|X) = \frac{1}{1+\alpha}$. Which of the following are true:
- (a) $\alpha = \frac{P(Y=0)P(X|Y=0)}{P(Y=1)P(X|Y=1)}$
 (b) It is possible to transform the Naive Bayes classifier to match the form of a logistic regression model, taking into account the assumptions given.
 (c) The class-conditional density being a Gaussian requires us to assume a Gaussian prior for Y
 (d) None of these
- 2(vii) Suppose we have observations $x_1, x_2, \dots, x_n$ of a continuous r.v. X known to be drawn from a “mixture” of k Gaussian distributions. Which of the following are true:
- (a) The number of parameters to be estimated is $2k$
 (b) $P(X = x_i) = P(x_i) = \sum_{j=1}^k P(x_i|j)P(j)$
 (c) $P(x_i|j) = \frac{1}{\sqrt{2\pi}\sigma_j} \exp\left(-\frac{(x_i - \mu_j)^2}{2\sigma_j^2}\right)$
 (d) None of these
- 2(viii) Assume we have image-data in which pixels are either black or white. Assume we will represent each image as d -dimensional Boolean vectors: each dimension represents a pixel, and a data-instance has the value 0 in the i^{th} dimension if the corresponding pixel is black, and 1 if the pixel is white. A simple generative model is one in which we assume each pixel in an image is independent of each other (yes, a bad assumption) and that each image is an instance of a d -dimensional random variable $\mathbf{X} = [X_1, X_2, \dots, X_d]^T$ in which of the $X_i \sim B(\theta_i)$, where $B(\theta)$ is univariate Bernoulli distribution with parameter θ . (Note: If $X \sim B(\theta)$ then $P(X = x|\theta) = \theta^x(1 - \theta)^{(1-x)}$ where $x \in \{0, 1\}$.) Which of the following are true:
 The result is a multivariate Bernoulli model $P(\mathbf{X}|\Theta)$ where $\Theta = \theta_1, \theta_2, \dots, \theta_d$. Which of the following are true:
- (a) if $X \sim B(\theta)$, then $E[X] = \theta$ and $Var[X] = \theta(1 - \theta)$
 (b) For a specific image $\mathbf{x} = [x_1, x_2, \dots, x_d]^T$, $P(\mathbf{x}|\Theta) = \prod_{i=1}^d \theta_i^{x_i} (1 - \theta_i)^{(1-x_i)}$
 (c) If we want to cluster the images into 2 groups then we can use a Gaussian mixture model to estimate the probability that an image belongs each of the 2 clusters.
 (d) None of these
- 2(ix) Consider the following figure:



Which of the following are correct:

- (a) The clusters can be found by a K-means procedure that incrementally tries $K = 2, 3, \dots$
 (b) The clustering could be found by a decision-tree algorithm with suitable data augmentation that adds “negative” examples

- (c) The clusters can be found by simply finding minimum and maximum X and Y values for points in each cluster, and using those to find the end-points of the rectangles
- (d) None of these
- 2(x) A search procedure for Bayesian networks scores networks using some weighted combination of the likelihood of the data and a prior over networks. If two BNs encode the same set of conditional independences, then they are said to be *Markov equivalent*. If networks g_1 and g_2 are Markov equivalent, and we are given some data d , then $P(d|g_1) = P(d|g_2)$. Which of the following are true:
- (a) Markov equivalent networks encode the same joint probability distribution
- (b) Markov equivalent networks have the same prior
- (c) The presence of Markov equivalent networks means the same network may not be returned by two different search procedures that use the same scoring function
- (d) None of these

Answer Here:

2(i)	
2(ii)	
2(iii)	
2(iv)	
2(v)	
2(vi)	
2(vii)	
2(viii)	
2(ix)	
2(x)	

[10 × 2 = 20 Marks]

Problem 3

3. Suppose you are interested in modelling the joint expression level of a pair of genes RAS2 and CS1 over a variety of physiological conditions. There are two hypotheses favoured in the biological literature: (a) H1: RAS2 regulates CS1; and (b) H2: RAS2 and CS1 are independent. Neither seems to be more likely, a priori. To resolve the matter, you have written an information-retrieval (IR) bot that extracted the following from scientific papers:
- | | |
|--|--|
| 1. “no evidence of RAS2 or CS1 levels” | 5. “although CS1 expression levels were high, RAS2 levels were not detected” |
| 2. “RAS2 and CS1 expression levels remain low” | 6. “RAS2 expression suppresses CS1” |
| 3. “RAS2 and CS1 levels were not significantly high” | 7. “RAS2 and CS1 were both expressed albeit not at the same time” |
| 4. “supports low CS1 levels causes low RAS2 levels” | 8. “RAS2 and CS1 levels are both high” |

- (a) Taking expression levels of genes simply as being “on” (1) or “off” (0), translate the data from the IR bot into a table of discretised values for RAS2 and CS1.

Statement	RAS2	CS1
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		

- (b) Translate the hypotheses H1 and H2 into Bayesian networks G1 and G2. Assume either expression is equally likely *a priori*, although the evidence for this is slightly stronger in the case of RAS2 than CS1, if the two are dependent. Specify the Bayesian network *completely*.

Bayesian network G1:

Bayesian network G2:

- (c) Which of the hypotheses H1 or H2 is better supported by the data? Show calculations needed to support your answer.

Calculation of $P(Data|H_1)$:

Calculation of $P(Data|H_2)$:

Final Conclusion:

[3 + 2 + 5 = 10 Marks]